

Laboratory Assignment 10: Counters
ECE 0201: Digital Circuits and Systems
(50 Points)

Name

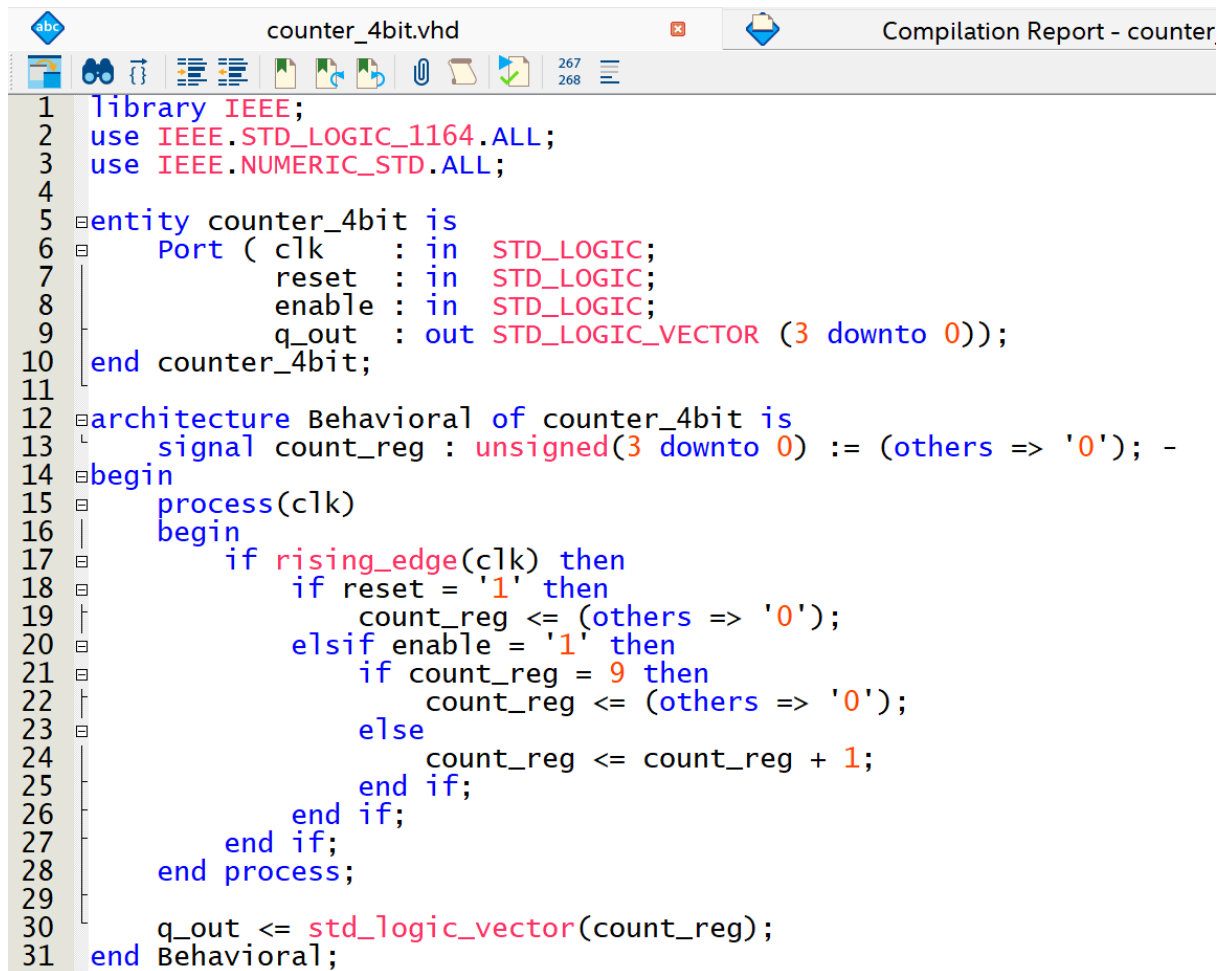
张鹤扬 Stefan Zhang

Submission Checklist:

- ☐ Write within boxes, do not move boxes
- ☐ Write your full name in the box above
- ☐ Save this file as a PDF before uploading, keep the number of pages (9) unchanged
- ☐ Note “TO BE CONTINUED” in the answer box if you used the extra pages (8-9)

Part I: Design of Synchronous Counters Using Behavioral HDL (5 points)

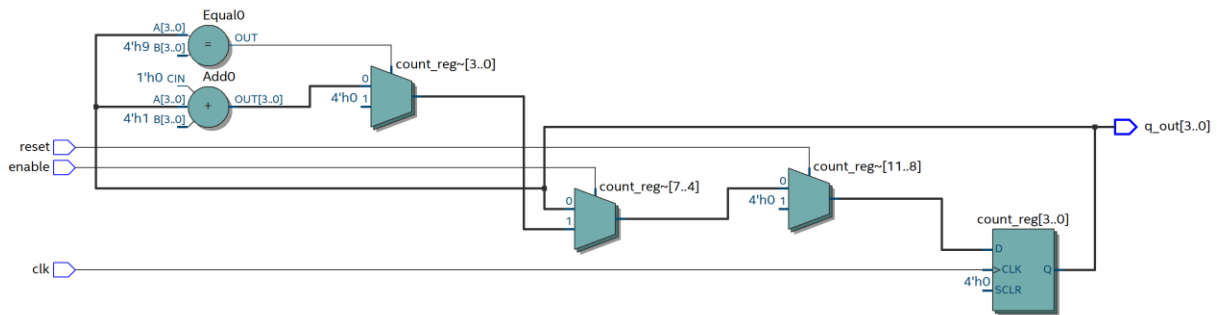
[(A) Insert a screenshot of your VHDL code for Part I.] (1 points)



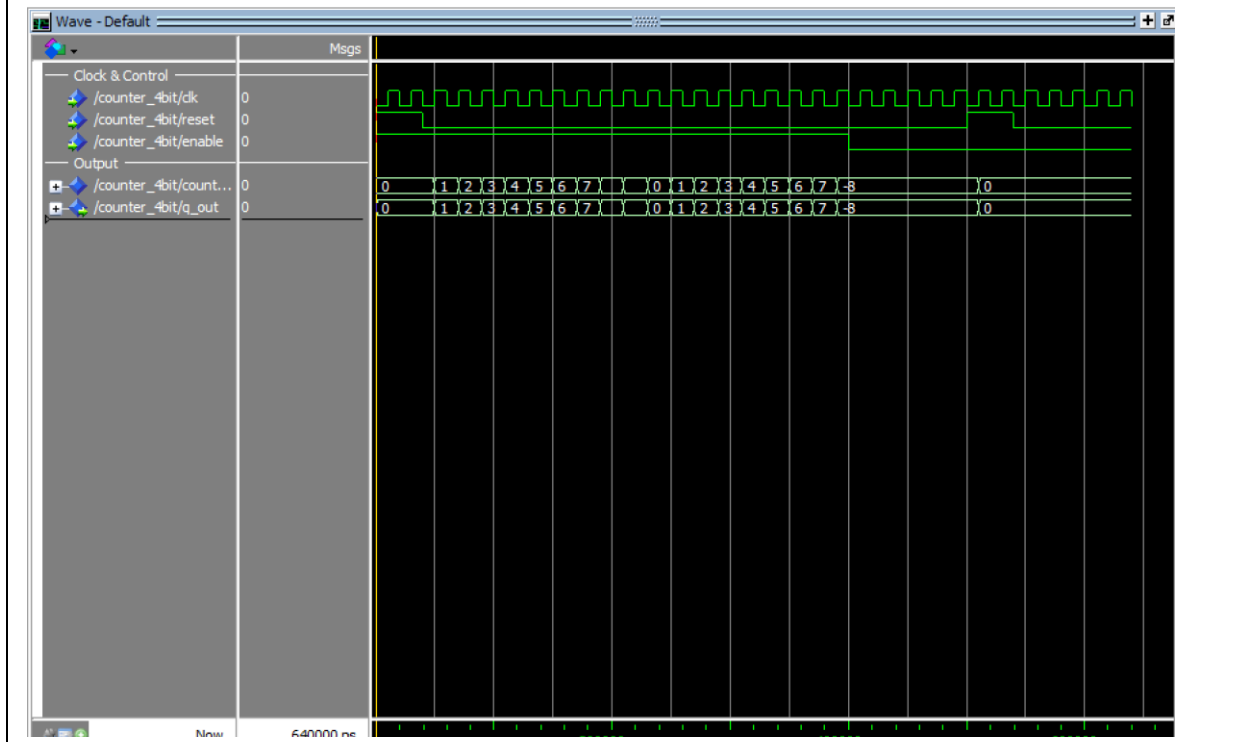
The screenshot shows a VHDL code editor window titled "counter_4bit.vhd". The code is as follows:

```
1  library IEEE;
2  use IEEE.STD_LOGIC_1164.ALL;
3  use IEEE.NUMERIC_STD.ALL;
4
5  entity counter_4bit is
6  Port ( clk      : in  STD_LOGIC;
7        reset    : in  STD_LOGIC;
8        enable   : in  STD_LOGIC;
9        q_out    : out STD_LOGIC_VECTOR (3 downto 0));
10 end counter_4bit;
11
12 architecture Behavioral of counter_4bit is
13     signal count_reg : unsigned(3 downto 0) := (others => '0'); -
14 begin
15     process(clk)
16     begin
17         if rising_edge(clk) then
18             if reset = '1' then
19                 count_reg <= (others => '0');
20             elsif enable = '1' then
21                 if count_reg = 9 then
22                     count_reg <= (others => '0');
23                 else
24                     count_reg <= count_reg + 1;
25                 end if;
26             end if;
27         end if;
28     end process;
29
30     q_out <= std_logic_vector(count_reg);
31 end Behavioral;
```

[(B) Insert a screenshot of your RTL Viewer schematic for Part I.] (2 points)

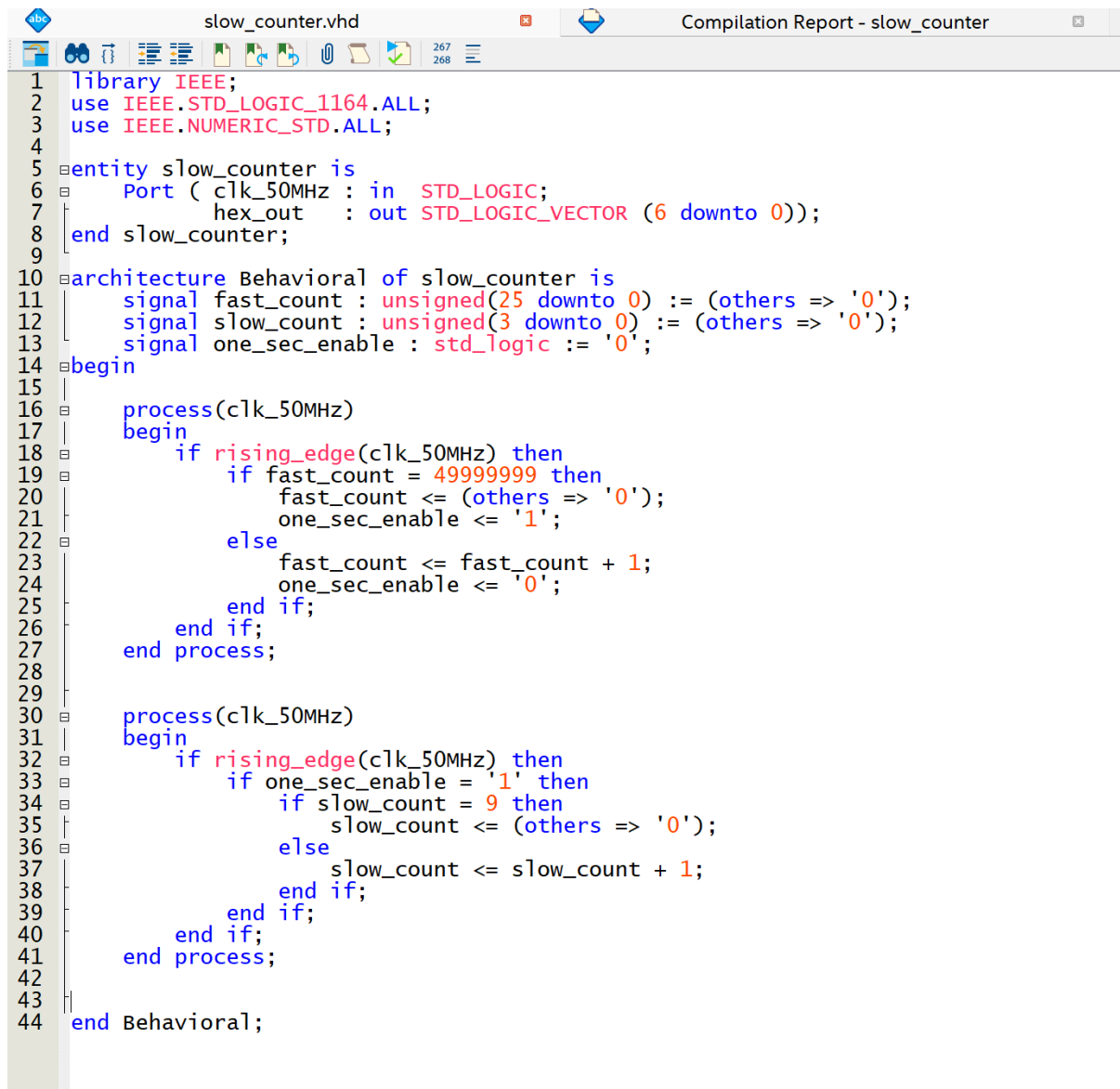


[(C) Insert a screenshot of your functional waveform simulation showing inputs and simulated outputs for at least 10 clock cycles.] (2 points)



Part II: Clocking Circuitry (20 points)

[(A) Insert a screenshot of your VHDL code for Part II.] (2 points)



```
1 library IEEE;
2 use IEEE.STD_LOGIC_1164.ALL;
3 use IEEE.NUMERIC_STD.ALL;
4
5 entity slow_counter is
6     Port ( clk_50MHz : in  STD_LOGIC;
7           hex_out   : out STD_LOGIC_VECTOR (6 downto 0));
8 end slow_counter;
9
10 architecture Behavioral of slow_counter is
11     signal fast_count : unsigned(25 downto 0) := (others => '0');
12     signal slow_count : unsigned(3 downto 0) := (others => '0');
13     signal one_sec_enable : std_logic := '0';
14 begin
15
16     process(clk_50MHz)
17     begin
18         if rising_edge(clk_50MHz) then
19             if fast_count = 4999999 then
20                 fast_count <= (others => '0');
21                 one_sec_enable <= '1';
22             else
23                 fast_count <= fast_count + 1;
24                 one_sec_enable <= '0';
25             end if;
26         end if;
27     end process;
28
29
30     process(clk_50MHz)
31     begin
32         if rising_edge(clk_50MHz) then
33             if one_sec_enable = '1' then
34                 if slow_count = 9 then
35                     slow_count <= (others => '0');
36                 else
37                     slow_count <= slow_count + 1;
38                 end if;
39             end if;
40         end if;
41     end process;
42
43 end Behavioral;
```

[(B) Demonstrate your circuit to an instructor or TA for the first Lab 10 check-off.] (18 points)

Part III: Rotating Word Display (5 points)

[(A) Insert a screenshot of your VHDL code for Part III.] (5 points)

Part IV: Augmenting the Rotating Display (20 points)

[(A) Insert a screenshot of your VHDL code for Part IV.] (2 points)

[(B) Demonstrate your circuit to an instructor or TA for the second Lab 10 check-off.] (18 points)

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